

SARVAGYA SINGH

Bokaro, India | singhsarvagya260508@gmail.com | saryagya.singh.me
[LinkedIn](#)

PROFILE

Building AI and software systems that work where infrastructure is thin. Most of my projects sit at the intersection of health, education, and agriculture in rural India and emerging markets. I ship systems that real people use: hospitals, farms, and schools. I am not interested in prototypes that stay on paper.

EDUCATION

Hong Kong University of Science and Technology	<i>B.Eng. Artificial Intelligence</i>	<i>Sept 2026 - 2030</i>
Delhi Public School, Bokaro (CBSE)	<i>Class XII SAT: 1460</i>	<i>2024 - 2026</i>
<i>Class X: 96.8% Class XII Prelim (AISSCE): 96.4%</i>		

PROFESSIONAL EXPERIENCE

Full Stack and DevOps Engineer	<i>Education India</i>	<i>May 2025 - Present</i>
- Building an edtech platform that helps students in India find courses, track certifications, and see their progress. Stack is React, Node.js, PostgreSQL. - Set up CI/CD with GitHub Actions and Docker. Cut deployment time by 40%. - Cut grading time from 12 seconds to under 2 seconds with Redis and Celery. Teachers actually use it now.		
Backend Engineer	<i>Education India</i>	<i>Apr 2024 - May 2025</i>
- Built the backend that processed 1,000+ student applications. Shortlisting, interviews, status tracking, the whole flow. - Built REST and GraphQL APIs for quizzes, assignments, and auto-grading. Added a vector DB so the platform can recommend relevant courses to each student. - Implemented role-based auth, rate limiting, and audit logging. Student data stays private and compliant.		
Software Engineering Intern	<i>Education India</i>	<i>Jan - Apr 2024</i>
- Built MVP features: form validation, document upload, and status updates. React and FastAPI. - Prototyped a quiz engine with question banks, timers, and auto-grading for MCQs. - Worked in agile sprints, handled Git workflows, and debugged production issues alongside the team.		

SELECTED HONOURS

- UNESCO Best Scientific Creativity Award.** Global top prize out of 3,062 submissions. Built a PDE-constrained PINN loss function and trained a photon transport simulator. [UNESCO feature.](#)
- Future Leader Award,** Hong Kong University of Science and Technology. Recognized for leading cross-functional AI deployment in resource-constrained settings.
- Global Rank 1,** ACES ISRC (1,000+ teams). Built a multimodal cardiac AI pipeline and a hospital validation protocol. [Live demo.](#)
- National Rank 2,** NIT Jamshedpur Conclave. Built edge-AIoT anomaly detection and deployed a telemedicine system. [Live demo.](#)
- All-India Rank 15,** WRO Nationals. Built a ROS2 navigation stack and real-time YOLO crack detection. Featured in 10+ national newspapers.
- IISER Bhopal Summer Research Fellow.** Top 7% of 3,500 applicants. Worked on experimental design and reproducibility frameworks for clinical AI validation.

RESEARCH AND PROJECTS

Pulsedage - Clinical AI for HCM	<i>Team Member (3-person team) pulsedage.vercel.app</i>	<i>2024</i>
- Built a multimodal CNN ensemble (ResNet-50 + EfficientNet-B4) that fuses ECG and MRI data for HCM detection. Picked CNN over ViT because hospital servers in India could not handle the compute load. - Built DICOM/HL7 preprocessing and used class-weighted focal loss to handle a 1:8 clinical prevalence imbalance. This matters when you are screening low-prevalence populations. - Tested at Bokaro General Hospital (500 beds). 98.5% sensitivity and specificity on 200+ patients. I handled the deployment and the feedback loop with clinicians. - Got inference under 200ms using INT8 quantization and Vercel edge. Runs on cheap hospital hardware with no GPU required.		
Nandani - Edge-AIoT Livestock Platform	<i>Lead Engineer (3-person team) gauseva-ai.vercel.app</i>	<i>2023 - 2024</i>
- Shrunk an LSTM model to fit on a Cortex-M4 collar with 256KB RAM. Runs under 10mW so it works 24/7 on a farm without draining the battery. - Built an offline-first MQTT and SQLite sync architecture. Farms in Jharkhand have spotty 2G at best, so reliable telemetry had to work without constant connectivity. - Deployed a low-bandwidth telemedicine system for 40+ smallholder farms in Jharkhand. Worked with IIT (ISM) on hardware prototyping. - Built a hardware-aware INT8 quantization pipeline. Cut false positives by 40%. Farmers catch diseases earlier and lose fewer animals, which protects their income.		
PromptBiotics - AI Nutrition Platform	<i>NUS Emergent Ventures (Top 0.4%) promptbiotics.org</i>	<i>Jun - Aug 2024</i>
- Built a RAG system with a fine-tuned GPT-4o. Hybrid semantic and BM25 retrieval, plus cross-encoder reranking. Had to balance relevance against latency. - Added a medical safety validation layer with constraint-based output filtering. Cut hallucinations by 60%. Recommendations stay within WHO dietary guidelines. - Beta launched to 500+ users. Tracked retention actively. Handled prompt engineering, safety evals, and the pitch to VCs. - Added a user feedback loop so prompts improve based on how people actually use the system.		
Autonomous Inspection Rover	<i>WRO Nationals - AIR 15</i>	<i>2024</i>
- Built a ROS2 rover with LiDAR and camera fusion for crack detection in hazardous zones. Picked EKF fusion over particle filter to keep latency low on edge GPUs. - A* and DWA path planning. TensorRT-accelerated YOLO at 35 FPS, under 50ms latency. Humans do not need to enter dangerous areas. - Led the navigation stack and CV integration. Coordinated sensor calibration, Gazebo simulation, and field testing with a team of three. - Documented the full ROS2 architecture. Featured in 10+ newspapers for making infrastructure inspection safer.		